FROM VECTORS TO QUANTUM WORLDS

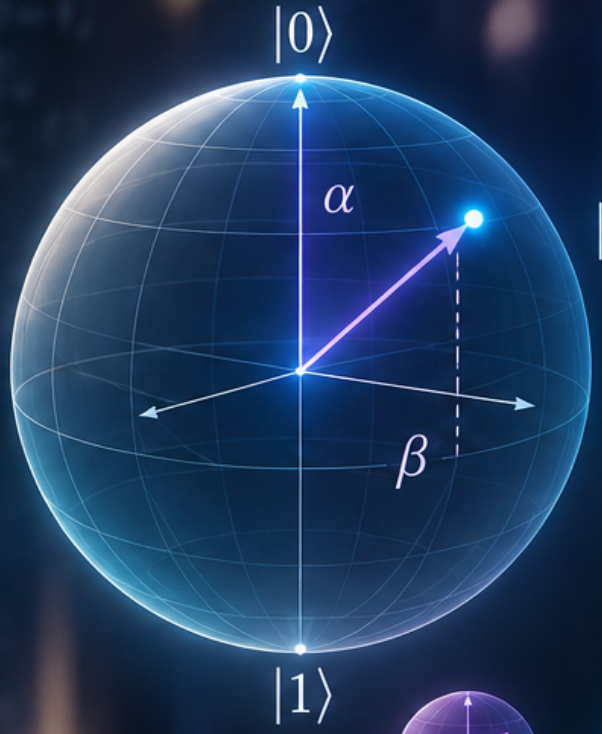
Linear Algebra *for* Quantum Computing

THE MATHEMATICS BEHIND
A NEW ERA OF COMPUTING

- $\begin{bmatrix}1\\0\end{bmatrix}$ **Vectors & Spaces**
  STATES AND SUPERPOSITION
- **Inner Products & Norms**
  MEASURING PROBABILITIES
- $[U]$ **Unitary & Hermitian Matrices**
  QUANTUM GATES AND OBSERVABLES
- λ **Eigenvalues & Eigenvectors**
  MEASUREMENT OUTCOMES
- **Tensor Products**
  MULTI-QUBIT SYSTEMS
- **Entanglement**
  BEYOND THE CLASSICAL WORLD

SAME MATH • BIGGER POSSIBILITIES

MATH
IDEAS
REAL IMPACT



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$U = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$$



$$= |\psi\rangle \otimes |\phi\rangle$$

$$|00\rangle \quad |01\rangle \quad |10\rangle \quad |11\rangle$$

2^n DIMENSIONS



BUILDING TOMORROW WITH
LINEAR ALGEBRA

Quantum Computing

ML/AL → CPU/GPU

Classical computer: Bits

0/1

Quantum Computing → qubits

$$|\psi| = \alpha|0| + \beta|1|$$

α, β → complex variable

P(0) P(1) states → undefined

$$|\alpha|^2 + |\beta|^2 = 1$$

$$|\alpha|^2 + |\beta|^2 = 1$$

idea

Quantum Computing ⟶ linear Algebra

IDEAS

1) Superposition ⟶ linear combination

2) Entanglement ⟶ Complex Vector

↳ Unfactorizable states

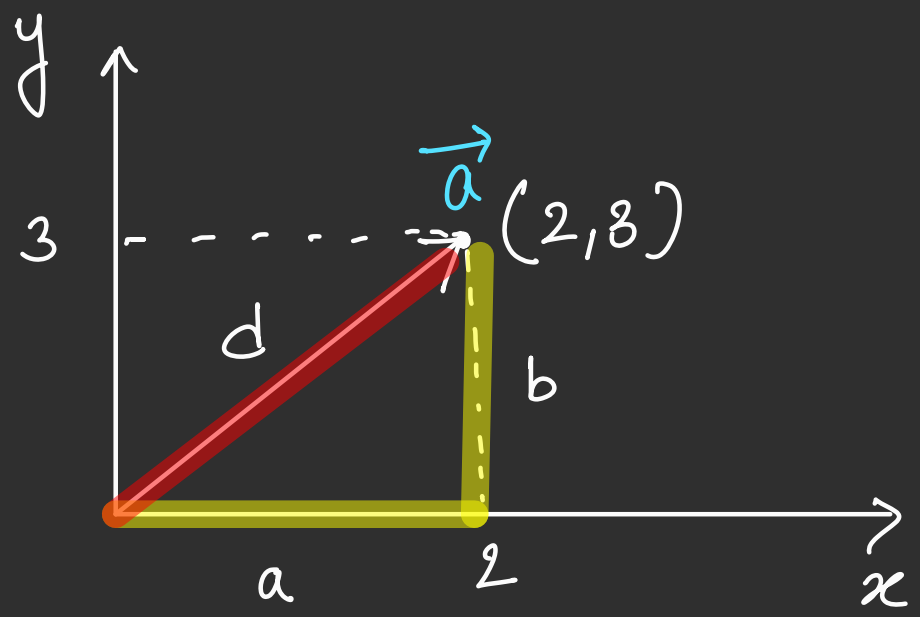
↳ Quantum State

Linear Algebra

phy & Math

Vector ⟶ Dir & Mag

$\vec{a}$ / $\underline{a}$



// Norm$_2$ ⟶ Euclidean dist

Norm$_1$ ⟶ Manhattan dist

$\vec{a}$ ⟶ (2,3) direc

$$d = \sqrt{a^2+b^2} = \sqrt{2^2+3^2} = \underline{\underline{\sqrt{13}}}$$

↳ pythagorous theorm

Mag ⟶ length of the Vector ⟶ (Norm)

$\|\vec{a}\|_2$ / $\|\vec{a}\|_1$

Norm$_1$ ⟶ $3+2 = \underline{\underline{5}}$

$$a = [1, 2]^T$$

Row & Cols → swap

$$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$v_1 \rightarrow [1, 3]$

$v_2 \rightarrow [4, 8]$

$a[1, 3] + b[4, 8]$

scalar values

$a\vec{v_1} + b\vec{v_2}$

$\text{span}\{v_1, v_2\}$

row vector

$$\vec{a} = [1 \ 2 \ 3]$$

col vector

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\bar{v} = [1\ 2\ 3]$

$\bar{u} = [4\ 5\ 6]$

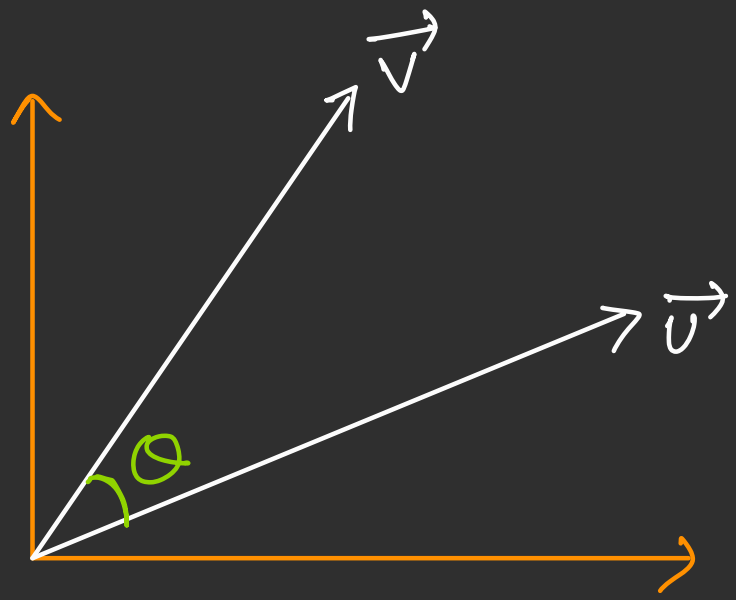
Mat mult

$[\ ]_{a \times \underline{b}}\ [\ ]_{\underline{b} \times c}$

$$\bar{v}^T \bar{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [4\ 5\ 6] = 4(1) + 5(2) + 6(3)$$

$$\Rightarrow 4 + 10 + 18 \Rightarrow \boxed{32\ \text{m}}$$

Angle b/w two vectors



Random Number

$\cos\theta = 0.34$

$\theta = \cos^{-1}(0.34)$

$$\cos\theta = \frac{\vec{v} \cdot \vec{u}}{\|\vec{v}\| \cdot \|\vec{u}\|}$$

$\vec{v} \cdot \vec{u}$ → dot product

$\|\vec{v}\| \cdot \|\vec{u}\|$ → multiplication of norm

$$\overline{x} = \begin{bmatrix} 1 \\ 3+2i \end{bmatrix}$$

$$\overline{x}^{*} = \begin{bmatrix} 1 \\ 3-2i \end{bmatrix}$$

$$\cos\theta = \frac{\vec{a}^T \cdot \vec{b}}{\|\vec{a}\| \, \|\vec{b}\|}$$

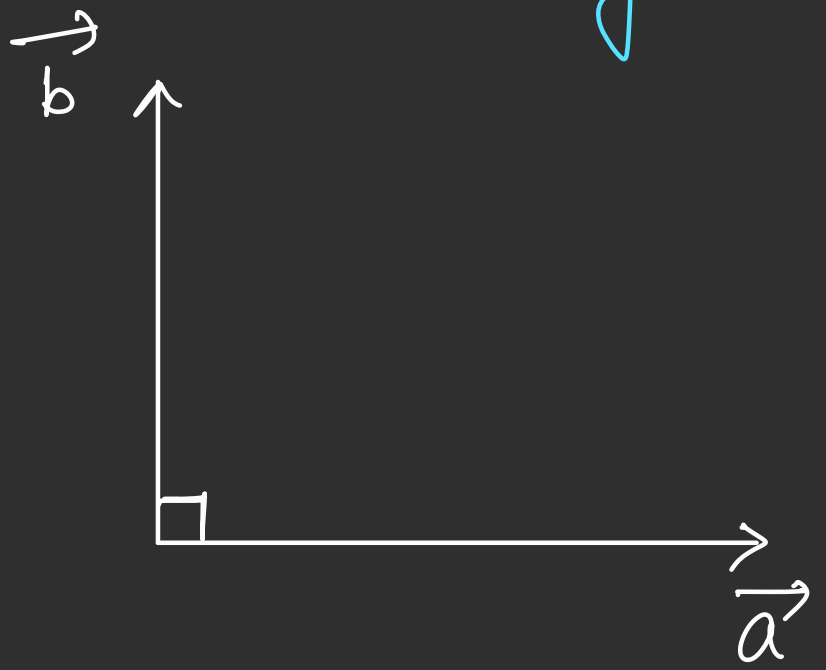
$\cos\theta = 0$

$\theta = 90^\circ$

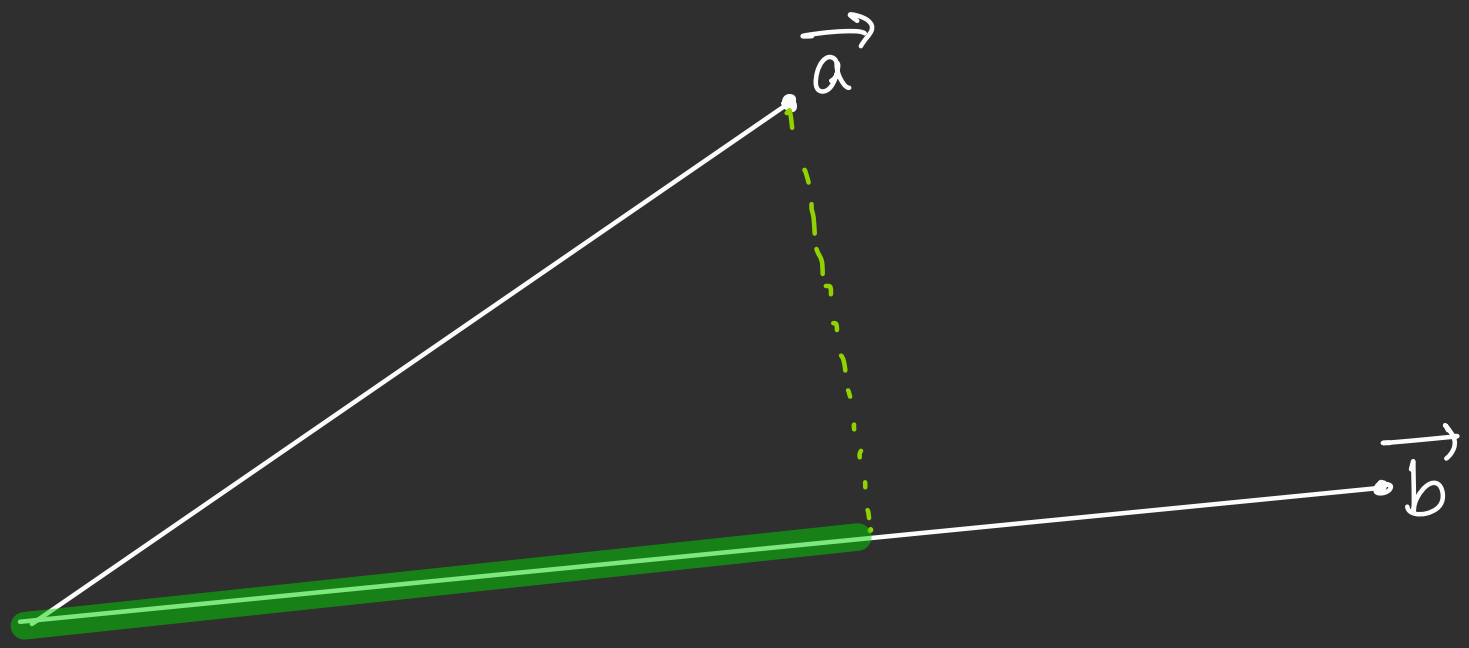
Num = 0

$\vec{a}^T \cdot \vec{b} = 0$

orthogonal vectors



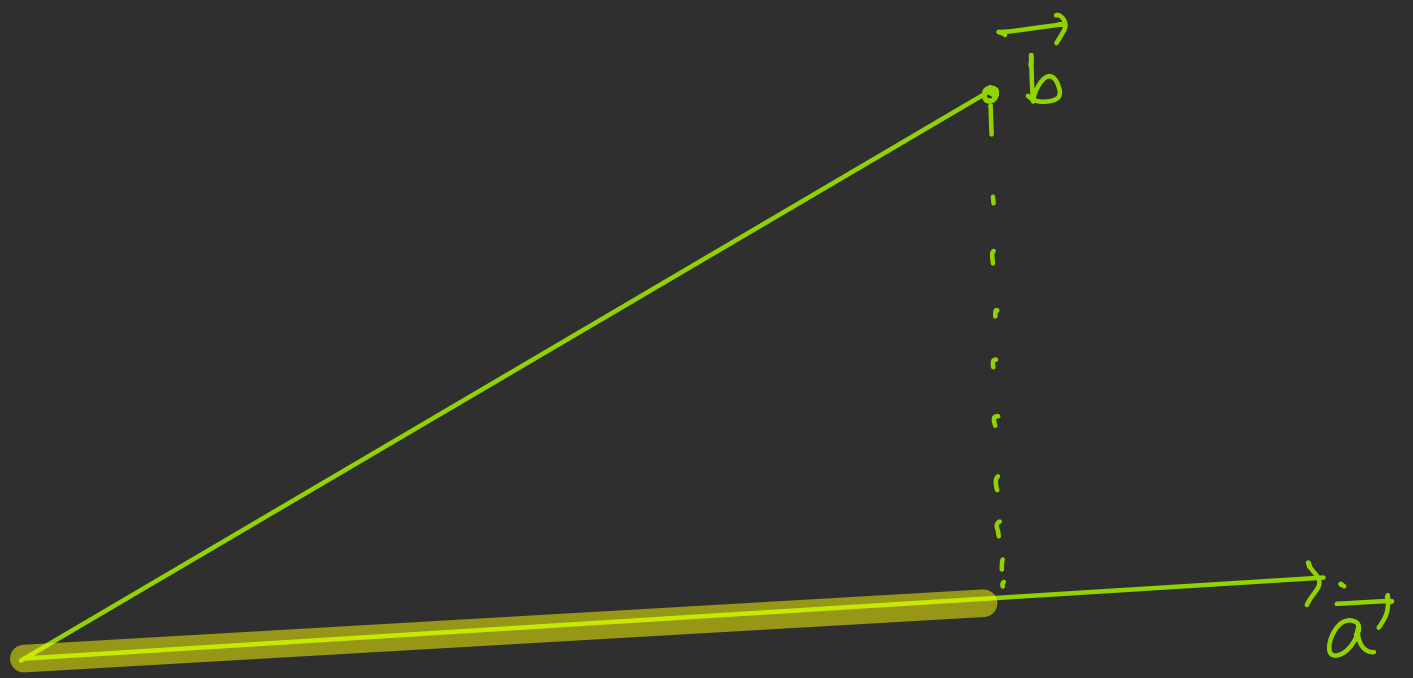
projection



projection of $\vec{a}$ on $\vec{b}$ $\Rightarrow$ $\dfrac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|} = \vec{a}\cdot\hat{b}$

$\hat{b}$ → unit vector

$(\|\hat{b}\| \to 1)$



Project $\vec{b}$ on $\vec{a}$ $\Rightarrow$ $\dfrac{\vec{b}\cdot\vec{a}}{\|\vec{a}\|} = \vec{b}\cdot\hat{a}$

$$A = \begin{bmatrix} 1 & 4 & 3 \\ 5 & 2 & 2 \\ 8 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

matrix

B = shape $\rightarrow (5, 2)$

B^T = shape $\rightarrow (2, 5)$

$$A^T = \begin{bmatrix} 1 & 5 & 8 \\ 4 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}_{3 \times 3}$$

$$A = \begin{bmatrix} \textcircled{1} & \textcircled{4} & \textcircled{3} \\ 5 & 2 & 2 \\ 8 & 1 & 1 \end{bmatrix}_{m \times n}^{3 \times 3}$$

$$A = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$H^T = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} \longrightarrow H^* = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Eigen Vectors & Values

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$Av = \lambda v$ — λ: eigen value; v: eigen vector

Find λ

$$\det(A - \lambda I) = 0$$

I: identity matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$(2-\lambda)(2-\lambda) - 1$$

$$(2-\lambda)^2 - 1 \Rightarrow 4 + \lambda^2 - 4\lambda - 1 = 0$$

$$\begin{cases} \lambda_1 = 1 \\ \lambda_2 = 3 \end{cases}$$

$$(A - \lambda I)v = 0$$

$$\lambda = 3$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

λ → 1

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + y = 0$$

$$x = -y$$

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$v_2 =$ ✗

Vectors ⟶ matrix ⟶ Tensor

$$a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

speed Vs accuracy

$$a \circledast b = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{bmatrix}$$

Vectors ⟶ Tensor products ⟶ multiple qubits ⟶ Entanglement